

*An Abstract for
a Research Paper*

Unit 7

Hongmei pang

This is the long and short of it.

– Shakespeare
Merry Wives of Windsor
Act II, scene ii

- Writing titles
- Writing abstracts
- Collocation



Titles

A title is to a research paper what a headline is to a news article!

The title should

- predict the content of the research paper.
 - topic
 - scope
 - reflect the tone of the writing.
 - contain important keywords.
 - be self-explanatory.
 - be interesting to the reader
- ✓ Two features: **informativity** and **economy**
- x “clever”, “jokes”, “trick” titles



Titles

Declarative titles

- state the main findings or conclusions
- Income moderates changes in big-five personality traits across eighteen years
- Morning employees are perceived as better employees: Employees' start times influence supervisor performance ratings
- I feel like I know you: Sharing negative attitudes of others promotes feelings of familiarity

Titles

Article | [Open access](#) | Published: 20 February 2023

Early morning university classes are associated with impaired sleep and academic performance

[Sing Chen Yeo](#), [Clin K. Y. Lai](#), [Jacinda Tan](#), [Samantha Lim](#), [Yuvan Chandramoghan](#), [Teck Kiang Tan](#) & [Joshua J. Gooley](#) 

[Nature Human Behaviour](#) **7**, 502–514 (2023) | [Cite this article](#)

Early morning university classes are associated with impaired sleep and academic performance

Attending classes and sleeping well are important for students' academic success. Here, we tested whether early morning classes are associated with lower attendance, shorter sleep and poorer academic achievement by analysing university students' digital traces. Wi-Fi connection logs in 23,391 students revealed that lecture attendance was about ten percentage points lower for classes at 08:00 compared with later start times. Diurnal patterns of Learning Management System logins in 39,458 students and actigraphy data in 181 students demonstrated that **nocturnal sleep was an hour shorter for early classes because students woke up earlier than usual**. Analyses of grades in 33,818 students showed that **the number of days per week they had morning classes was negatively correlated with grade point average**. These findings suggest concerning associations between early morning classes and learning outcomes.

Sleepmore in Seattle: Later school start times are associated with more sleep and better performance in high school students

Gideon P. Dunster¹, Luciano de la Iglesia¹, Miriam Ben-Hamo¹, Claire Nave¹, Jason G. Fleisc...

✚ See all authors and affiliations

Science Advances 12 Dec 2018:
Vol. 4, no. 12, eaau6200
DOI: 10.1126/sciadv.aau6200

Abstract

Most teenagers are chronically sleep deprived. One strategy proposed to lengthen adolescent sleep is to delay secondary school start times. This would allow students to wake up later without shifting their bedtime, which is biologically determined by the circadian clock, resulting in a net increase in sleep. So far, there is no objective quantitative data showing that a single intervention such as delaying the school start time significantly increases daily sleep. The Seattle School District delayed the secondary school start time by nearly an hour. We carried out a pre-/post-research study and show that there was an increase in the daily median sleep duration of 34 min, associated with a 4.5% increase in the median grades of the students and an improvement in attendance.

Titles

Descriptive titles

- describe the subject of the article but do not reveal the main conclusions
- Effect of increased gravitational acceleration in potato deep-fat frying
- CEO early-life disaster experience and corporate social performance
- The influence of environmental and social characteristics on women's mate preferences

Titles

Journal List > J Appl Behav Anal > v.7(3); Fall 1974 > PMC1311997

JOURNAL OF APPLIED BEHAVIOR ANALYSIS

Society for the Experimental Analysis of Behavior

[J Appl Behav Anal](#). 1974 Fall; 7(3): 497.

doi: [10.1901/jaba.1974.7-497a](#)

PMCID: PMC1311997

PMID: [16795475](#)

The unsuccessful self-treatment of a case of "writer's block"¹

[Dennis Upper](#)

JOURNAL OF APPLIED BEHAVIOR ANALYSIS 1974, 7, 497 NUMBER 3 (FALL 1974)

THE UNSUCCESSFUL SELF-TREATMENT OF A CASE OF "WRITER'S BLOCK"¹

DENNIS UPPER

VETERANS ADMINISTRATION HOSPITAL, BROCKTON, MASSACHUSETTS

REFERENCES

¹Portions of this paper were not presented at the 81st Annual American Psychological Association Convention, Montreal, Canada, August 30, 1973. Reprints may be obtained from Dennis Upper, Behavior Therapy Unit, Veterans Administration Hospital, Brockton, Massachusetts 02401.

*Received 25 October 1973.
(Published without revision.)*

COMMENTS BY REVIEWER A

I have studied this manuscript very carefully with lemon juice and X-rays and have not detected a single flaw in either design or writing style. I suggest it be published without revision. Clearly it is the most concise manuscript I have ever seen—yet it contains

sufficient detail to allow other investigators to replicate Dr. Upper's failure. In comparison with the other manuscripts I get from you containing all that complicated detail, this one was a pleasure to examine. Surely we can find a place for this paper in the Journal—perhaps on the edge of a blank page.

Titles

Interrogative titles

- introduce the subject in the form of a question
- How does the beauty of wives affect post-marriage family outcomes? Helen's face in Chinese households
- Why East Asians but not South Asians are underrepresented in leaderships positions in the United States?
- Does a long-term relationship kill romantic love?

Bullshitters. Who are they and what do we know about their lives?

[John Jerrim](#), [Phil Parker](#), [Nikki Shure](#)

'Bullshitters' are individuals who claim knowledge or expertise in an area where they actually have little experience or skill. Despite this being a well-known and widespread social phenomenon, relatively few large-scale empirical studies have been conducted into this issue. This paper attempts to fill this gap in the literature by examining teenagers' propensity to claim expertise in three mathematics constructs that do not really exist. Using Programme for International Student Assessment (PISA) data from nine Anglophone countries and over 40,000 young people, we find substantial differences in young people's tendency to **bullshit across countries, genders and socio-economic groups**. Bullshitters are also found to exhibit high levels of overconfidence and believe they work hard, persevere at tasks, and are popular amongst their peers. Together this provides important new insight into who bullshitters are and the type of survey responses that they provide.

This paper has attempted to explore such issues using large-scale, nationally representative data. Focusing upon 15-year-olds from across nine Anglophone countries, we have investigated the characteristics of young people who claim to have knowledge and expertise in three mathematics concepts which are fake. Having derived and established the comparability of our bullshit scale via measurement invariance procedures, we go on to find that **young men are more likely to bullshit** than young women, and that bullshitting is somewhat **more prevalent amongst those from more advantaged socioeconomic backgrounds**. Compared to other countries, young people **in North America** are found to be bigger bullshitters than young people in England, Australia and New Zealand, while those in Ireland and Scotland are the least likely to exaggerate their mathematical knowledge and abilities. Strong evidence also emerges that bullshitters also display overconfidence in their academic prowess and problem-solving skills, while also reporting higher levels of perseverance when faced with challenges and providing more socially desirable responses than more truthful groups.

The grass is always greener on the other side.



When it comes to pay, do the thin win? The effect of weight on pay for men and women

Timothy A Judge ¹, Daniel M Cable

Abstract

Cultivation theory suggests that society holds very different body standards for men versus women, and research indicates that the consequences of defying these social norms may not be linear. To test these notions in the employment context, we examined the relationship between weight and income and the degree to which the relationship varies by gender. **For women**, we theorized **a negative weight–income relationship that is steepest at the thin end of the distribution**. For men, we predicted a positive weight–income relationship until obesity, where it becomes negative. To test these hypotheses, we utilized 2 longitudinal studies, 1 German and 1 American. In Study 1, weight was measured over 2 time periods, and earnings were averaged over the subsequent 5 years. Study 2 was a multilevel study in which weight and earnings were within-individual variables observed over time, and gender was a between-individual variable. **Results from the 2 studies generally support the hypotheses**, even when examining within-individual changes in weight over time.

Grammatical Constructions In Titles

- **full-sentence constructions**

Shhhh! Chinese students are studying quietly in the UK

- **nominal group constructions**

Acute liver failure caused by diffuse hepatic melanoma infiltration;

- **compound constructions** (i.e. divided into two parts, mainly by a colon)

Romanian nominalizations: case and aspectual structure

- **question constructions**

Does the Flynn effect affect IQ scores of students classified as learning-disabled?

The average percentage occurrence of title formats for research papers

	The sciences	The social sciences
Nominal group construction	52	61
Full-sentence construct	38	0.1
Compound construction	10	38
Question	0	0.3

Title formula

➤ [Topic]: [Method]

- Superstition and risk taking: Evidence from “zodiac year” beliefs in China
- Exploring university students’ perceptions of plagiarism: A focus group study
- The impact of Phd studies on mental health: A longitudinal population study

➤ [Conclusion]: [Results]

- Give me a straight answer: Responses ambiguity diminishes likability
- Give a dog a bone: Spending money on pets promotes happiness

➤ [Results]: [Method]

- The boredom proneness levels of Chinese college students increased over time: A meta-analysis of birth cohort differences from 2009 to 2020

➤ [Results]: A [method] study of [topic] among [sample]

- Meditation makes nurses perform better: A qualitative study of mindfulness meditation among Japanese nursing students

running head/running title /short title

- Shortened form of the main title
- Requirements for running titles vary between journals, but generally, they must be 50–60 characters long at most, often including spaces.
- To achieve brevity, these titles typically include abbreviations, even if the main title does not (or cannot, based on journal guidelines). Articles (the, a, an) may also be omitted to conserve characters, and wordy phrasing should be minimized.
- Clarity and accuracy should be priorities.
 - **Manuscript title:** Dendritic Cell Immunoreceptor Is a New Target for Anti-AIDS Drug Development: Identification of DCIR/HIV-1 Inhibitors
 - **Running head:** Inhibitors of DCIR Limit HIV-1 Infection (Lambert et al., 2013):

APA title page

Running head: VARYING DEFINITIONS OF ONLINE COMMUNICATION

1

The title should summarize the paper's main idea and identify the variables under discussion and the relationship between them.



Varying Definitions of Online Communication and
Their Effects on Relationship Research

Elizabeth L. Angeli
Purdue University

The author's name and institution should be double-spaced and centered.



The title should be centered on the page, typed in 12-point Times New Roman Font. It should not be bolded, underlined, or italicized.

The running head is a shortened version of the paper's full title, and it is used to help readers identify the titles for published articles (even if your paper is not intended for publication, your paper should still have a running head).

The running head cannot exceed 50 characters, including spaces and punctuation. The running head's title should be in capital letters. The running head should be flush left, and page numbers should be flush right. On the title page, the running head should include the words "Running head." For pages following the title page, repeat the running head in all caps without "Running head."

Blue boxes contain directions for writing and citing in APA style.

Green text boxes contain explanations of APA style guidelines.

Running head: THE EFFECTS OF CARBON-DATING TECHNOLOGY

1

The Effects of Carbon-Dating Technology on Dinosaur Bones

Jane Smith

ENG 122: English Composition II

Prof. James Lee

January 1, 2018

I was Young and Needed the Publication

That's me!

I had the Idea and had
to defend it against my advisor

Random Guy

That showed up
last minute

Never met

But important for
future funding

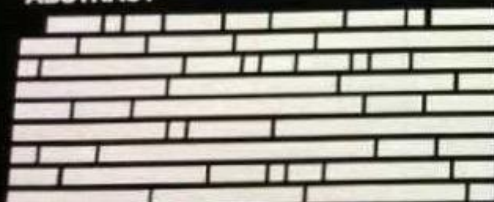
The Intern

That did all the
hard lab work

Advisor

He might have seen
the paper once or twice

ABSTRACT



- 老太太买鸡蛋

《论社会主义市场经济条件下高龄老嫗如何完成鸟卵与货币的置换》

—北京大学新闻与传播学院 陈汝东

科研有黑话，拒绝黑化会挥发

Abstract

- Results-Driven abstract
- Research paper summary-Driven abstract

Abstract 1



Journal of Economic Behavior &
Organization

Volume 150, June 2018, Pages 311-324

Title length

Yann Bramoullé ^a, Lorenzo Ductor ^b✉

We document **strong and robust negative correlations** between the length of the title of an economics article and different measures of scientific quality. Analyzing all articles published between 1970 and 2011 and referenced in EconLit, **we find that articles with shorter titles tend to be published in better journals, to be more cited and to be more innovative.**

Over the past few decades, **neuroimaging** has become a ubiquitous tool in basic research and clinical studies of the human brain. **However, no reference standards currently exist** to quantify individual differences in neuroimaging metrics over time, in contrast to growth charts for anthropometric traits such as height and weight. **Here we assemble an interactive open resource to benchmark brain morphology** derived from any current or future sample of MRI data. With the goal of basing these reference charts on the largest and most inclusive dataset available, acknowledging limitations due to known biases of MRI studies relative to the diversity of the global population, we **aggregated 123,984 MRI scans, across more than 100 primary studies, from 101,457 human participants between 115 days post-conception to 100 years of age**. MRI metrics were quantified by centile scores, relative to non-linear trajectories of brain structural changes, and rates of change, over the lifespan. **Brain charts identified previously unreported neurodevelopmental milestones, showed high stability** of individuals across longitudinal assessments, and demonstrated robustness to technical and methodological differences between primary studies. **Centile scores showed increased heritability compared with non-centiled MRI phenotypes, and provided a standardized measure of atypical brain structure that revealed patterns of neuroanatomical variation across neurological and psychiatric disorders**. In summary, brain charts are an **essential step towards** robust quantification of individual variation benchmarked to normative trajectories in multiple, commonly used neuroimaging phenotypes.

topic under study

gap

purpose

Methods

main results

Significance or implication

An abstract for an empirical study is typically written using five sub-headings

1. Background (1-2 sentences) (optional)
2. Aim / objective (obligatory)
3. Method (obligatory)
 - Participants
 - Techniques of data collection
4. Results (obligatory)
5. Discussion (obligatory)

Annotated example taken from *Nature* 435, 114–118 (5 May 2005).

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Two to three sentences of **more detailed background**, comprehensible to scientists in related disciplines.

One sentence clearly stating the **general problem** being addressed by this particular study.

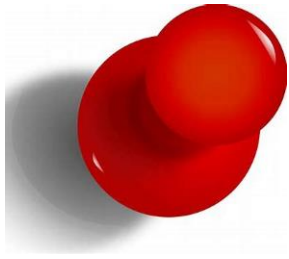
One sentence summarizing the main result (with the words “**here we show**” or their equivalent).

Two or three sentences explaining what the **main result** reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.

One or two sentences to put the results into a more **general context**.

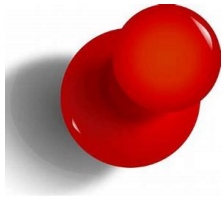
Two or three sentences to provide a **broader perspective**, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (This example is 190 words without the final section, and 250 words with it).

During cell division, mitotic spindles are assembled by microtubule-based motor proteins^{1,2}. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homotetrameric motor proteins of the widely conserved kinesin-5 (BimC) family³. Hypotheses for bipolar spindle formation include the ‘push–pull mitotic muscle’ model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules^{2,4,5}. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the vertebrate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled *in vitro* assays that Eg5 has the remarkable capability of simultaneously moving at $\sim 20 \text{ nm s}^{-1}$ towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at $\sim 40 \text{ nm s}^{-1}$, comparable to spindle pole separation rates *in vivo*⁶. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated *in vitro* models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.



- This report examines the causes of oversleeping.
- Individuals oversleep because they go to bed too late, forget to set their alarms, and keep their rooms dark.

a statement of what the paper will ask or explore VS what it found



- The study draws conclusions about which variables are most important in choosing a movie theater.
- The study concludes that the most important variables in choosing a movie theater are comfortable seats and high-quality popcorn.

general categories VS specific details in the findings

Structured abstract

- A structured abstract is an abstract with distinct, labeled sections.

Various Leisure-Time Physical Activities Associated With Widely Divergent Life Expectancies: The Copenhagen City Heart Study

Objective

To evaluate the differential improvements in life expectancy associated with participation in various sports.

Patients and Methods

The Copenhagen City Heart Study (CCHS) is a prospective population study that included detailed questionnaires regarding participation in different types of sports and leisure-time physical activity. The 8577 participants were followed for up to 25 years for all-cause mortality from their examination between October 10, 1991, and September 16, 1994, until March 22, 2017. Relative risks were calculated using Cox proportional hazards models with full adjustment for confounding variables.

Results

Multivariable-adjusted life expectancy gains compared with the sedentary group for different sports were as follows: tennis, 9.7 years; badminton, 6.2 years; soccer, 4.7 years; cycling, 3.7 years; swimming, 3.4 years; jogging, 3.2 years; calisthenics, 3.1 years; and health club activities, 1.5 years.

Conclusion

Various sports are associated with markedly different improvements in life expectancy. Because this is an observational study, it remains uncertain whether this relationship is causal. Interestingly, the leisure-time sports that inherently involve more social interaction were associated with the best longevity—a finding that warrants further investigation.

- Structured abstracts have several advantages for authors and readers.
- These formats were developed in the late 1980s and early 1990s to assist health professionals in selecting clinically relevant and methodologically valid journal articles.
- They also guide authors in summarizing the content of their manuscripts precisely, facilitate the peer-review process for manuscripts submitted for publication, and enhance computerized literature searching.

Typical headings

Background / Context / Purpose – Methods – Results / Findings – Conclusions

Context – Aim / Objective – Design – Setting – Patients (or Participants) – Interventions / Treatment – Main Outcome Measure(s) – Results – Conclusions

Context – Objective – Data Sources – Study Selection – Data Extraction – Results – Conclusions

FAST TRACK COMMUNICATION

Can apparent superluminal neutrino speeds be explained as a quantum weak measurement?

M V Berry¹, N Brunner¹, S Popescu¹ and P Shukla²

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² Department of Physics, Indian Institute of Technology, Kharagpur, India

Received 12 October 2011, in final form 27 October 2011

Published 11 November 2011

Online at stacks.iop.org/JPhysA/44/492001

Abstract

Probably not.

PACS numbers: 03.65.Ta, 03.65.Xp, 14.60.Pq



Rewrite the abstract

Tsinghua Students' Perceptions of Dictionary Use in English Vocabulary Learning

Abstract

Dictionaries are widely used in language learning. Students often look up dictionaries when they meet words they hadn't learned before. The relationship between dictionary use and vocabulary learning have been discussed for many times. Tsinghua, the best university in China, has many English learners. This research focuses on the Tsinghua students' perceptions of dictionary use in English vocabulary learning. The result which based on the interviews of 8 Tsinghua students shows the participants often preferred to use English-Chinese dictionaries, paid more attention to word meanings than word usage, and weren't willing to use paper dictionaries when learning English vocabulary. Tsinghua students should use more strategies to learn vocabulary with dictionaries.

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too general

collocation

contraction

tense

agreement

vague information

colloquial expression

Irrelevant information

“the” article

sentence structure

spell

non-affirmative

sentence connector

Dictionaries are widely used in language learning. Students often look up dictionaries when they meet words they hadn't learned before. The relationship between dictionary use and vocabulary learning have been discussed for many times. Tsinghua, the best university in China, has many English learners. This research focuses on the Tsinghua students' perceptions of dictionary use in English vocabulary learning. The result which based on the interviews of 8 Tsinghua students shows the participants often preferred to use English-Chinese dictionaries, paid more attention to word meanings than word usage, and weren't willing to use paper dictionaries when learning English vocabulary. Tsinghua students should use more strategies to learn vocabulary with dictionaries.

Dictionary use has long been recognized as one of vocabulary learning strategies. This research aims to explore Tsinghua students' perceptions of dictionary use in English vocabulary learning. The results, based on the interviews of eight Tsinghua students, show that the participants tend to use English-Chinese dictionaries, pay more attention to word meanings than word usage, and are unwilling to use paper dictionaries when learning English vocabulary. The study suggests that Tsinghua students should employ more diverse dictionary strategies to learn vocabulary.



Reading research papers



1. Introduction

- **Problem:** Students entering university often lack knowledge of academic writing
- **Context:**
 - Writing instruction: inconsistent/patchy
 - Existing support focuses on genre and skills rather than the cognitive processes involved in writing.
- **Gap in research:** Little is known about students' cognitive processes during writing.
- **Existing research:** Experts spend more time on planning and reviewing, while novices focus on formulation.
- **Limitations of existing research:**
 - Many studies focus on L2 writers, where proficiency level influences writing processes.
 - Studies often use a “one-shot” approach, which is not representative of academic writing.
 - Studies of naturalistic academic writing lack a comparison of process use with writing performance.
- **Methodological limitations:**
- **Research aim:** To compare the writing processes of successful and unsuccessful novice academic writers.

2. The Cognitive Process Model

- **Theoretical and analytical framework: Flower and Hayes (1981):**
 - Planning
 - Translating (formulating)
 - Reviewing (evaluation+revision)

interact recursively
- **Expert vs. novice writers:**
 - Experts : more likely to set precise goals, modify them throughout the writing process, and engage in more effective retrieval, development, and revision

3. Previous Process-Oriented Writing Research

- **Focus on L2 writing**
- **Time allocation patterns:**
 - More proficient writers spend more time on planning and reviewing, while less proficient writers focus on formulation.
- **Lack of focus on source use**
- **The digital environment**

4. Methodologies of Previous Process Studies

- **Methodological limitations:**
 - Think-aloud protocols
 - Retrospection

5. Research Aim

- **Aim:**
 - To compare the use of writing processes between successful and unsuccessful novice academic writers.
- **Research Questions:**
 - 1.What are the differences in time allocation to various writing processes between a successful and an unsuccessful novice academic writer?
 - 2.How do the writing behaviors differ between a successful and an unsuccessful novice academic writer?

6. Methods

- **6.1 Participants:**

- 13 students who were about to write their first assignment at university, from Australia and the UK.
- Ages ranging from 18 to 24, with only one non-native English speaker.
- Disciplines: Applied Linguistics, Political Science, Tourism & Event Management, and Chemistry.
- Genres: essays, a business report, and a literature review.
- Selection for case study: GINA (high-achieving) and ELLEN (low-achieving) based on complete screen recordings.

6. Methods

- **6.2 Data Collection:**
 - **Student Observation:**
 - Screen recording
 - Unobtrusive
 - **Student Interviews:**
 - Three interviews: planning stage, writing stage, and reflection stage.
 - Questions about writing experience, expectations, initial plans, writing process, and reflections on grades and feedback
 - Complements screen recordings
- **6.3 Data Analysis:**
 - **Quantitative Analysis:**
 - Adapted Flower and Hayes (1981)
 - Excel spreadsheets for coding and calculating time allocation to each sub-process
 - Analysis of overall time spent on each process and process interaction
 - **Qualitative Analysis:**
 - Verbal descriptions
 - Thematic analysis

7. Findings and Discussion

- **Time Allocation**
- **Planning:**
 - GINA : a significant portion of her time to planning, developing a detailed outline and structure for her essay.
 - ELLEN : minimal time planning and did not create a formal outline
- **Composing and Reviewing:**
 - GINA : less time on the actual writing process, significantly more time to reviewing and revising her draft
 - ELLEN: the majority of her time on composing and editing, little attention to revising

7. Findings and Discussion

- **Differences in Writing Behaviors**
- **Process Interaction:**
 - GINA : a strong interaction between composing and reviewing processes
 - ELLEN: limited process interaction
- **Recursiveness in Argument Development:**
 - GINA: a recursive approach to argument development, coherent argument
 - ELLEN: little consideration for the development or coherence, unorganized argument

8. Conclusion

- **Major Findings:**
 - Successful academic writers allocate more time to planning and reviewing.
 - Successful writers engage in more process interaction and recursiveness in argument development.
 - Unsuccessful writers tend to focus on composing and editing without adequate planning or revision.
- **Pedagogical Implications:**
 - Novice writers need to learn the importance of planning and reviewing.
 - Teaching should go beyond genre-based approaches and include reading-to-write activities.
 - Process-oriented instruction can help novice writers develop effective writing strategies.
 - Screen recordings can serve as a teaching tool.
- **Limitations:**
 - small sample



Work on the collocations.

- ① 勉强及格
- ② 花时间在...
- ③ 提供对信息源的即时访问
- ④ 即将到来的任务
- ⑤ 显著差异

- ① a bare pass
- ② spend... on/devote ... to/allocate ... to
- ③ afford immediate access to sources
- ④ the impending assignment
- ⑤ stark/considerable/sharp/significant difference

Learning achievements

- ✓ Writing titles
- ✓ Writing abstracts
- ✓ Collocation

